Supplementary Material to CPOR: Cumulative Protection for Online Reconciliation with Nodewise Error Limits

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Supplementary material for Information Sciences

S1 Accuracy and training-run variation

Tables S1–S6 report means and sample SDs across the three backbone training runs. Window entries first average the two window RMSEs within a run. Continuous entries use all 144 observations directly.

Table S1: Window RMSE with PatchTST.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1080.38 ± 27.08	1307.15 ± 40.97	1076.05 ± 55.04	1077.37 ± 55.71	948.66 ± 23.19	$\mathbf{947.27 \pm 23.82}$
ECommerce	1509.91 ± 1.86	1500.53 ± 7.82	1582.66 ± 15.89	1582.66 ± 15.89	1476.19 ± 4.58	$\mathbf{1470.64 \pm 3.52}$
Traffic	5.296 ± 0.143	3.841 ± 0.017	3.748 ± 0.162	$\mathbf{3.660 \pm 0.114}$	3.739 ± 0.030	3.743 ± 0.032
Wiki2	3172.20 ± 62.32	2546.76 ± 13.36	2772.26 ± 31.21	2872.75 ± 37.17	$\mathbf{2465.83 \pm 5.55}$	2483.48 ± 8.02
Labour	23.43 ± 0.43	21.07 ± 0.69	15.19 ± 0.36	15.46 ± 0.42	$\mathbf{14.81 \pm 0.06}$	$\mathbf{14.81 \pm 0.06}$
M5	429.56 ± 2.64	437.70 ± 3.91	417.04 ± 28.87	$\mathbf{408.00 \pm 26.94}$	421.62 ± 0.57	420.12 ± 0.46
AEMO	647.72 ± 62.65	668.35 ± 62.06	$\mathbf{474.21 \pm 25.35}$	481.64 ± 25.94	628.19 ± 61.00	629.02 ± 61.42

Table S2: Window RMSE with DLinear.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1459.30 ± 28.28	1901.25 ± 21.37	1238.61 ± 107.06	1241.07 ± 107.86	$\mathbf{1192.97 \pm 5.59}$	1193.89 ± 8.11
ECommerce	1562.45 ± 6.09	1541.79 ± 5.87	1617.31 ± 41.77	1617.31 ± 41.77	1494.24 ± 3.46	$\mathbf{1484.68 \pm 2.45}$
Traffic	5.365 ± 0.075	4.131 ± 0.055	3.864 ± 0.153	$\mathbf{3.855 \pm 0.120}$	3.977 ± 0.037	3.981 ± 0.040
Wiki2	3363.83 ± 60.72	2708.30 ± 17.60	2899.81 ± 60.41	3004.38 ± 51.77	$\mathbf{2548.73 \pm 4.70}$	2557.01 ± 9.66
Labour	25.61 ± 0.18	23.58 ± 0.58	18.57 ± 0.55	18.37 ± 0.56	$\mathbf{17.68 \pm 0.36}$	17.69 ± 0.36
M5	465.32 ± 16.89	468.52 ± 12.62	427.30 ± 30.02	$\mathbf{419.79 \pm 27.30}$	447.25 ± 10.57	445.95 ± 10.23
AEMO	1177.44 ± 12.12	1181.48 ± 17.04	646.44 ± 18.17	$\mathbf{645.09 \pm 18.42}$	806.56 ± 5.35	806.52 ± 5.39

Table S3: Window RMSE with DUET.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1142.72 ± 11.18	1529.48 ± 36.50	$\mathbf{947.28 \pm 36.74}$	948.40 ± 38.12	953.06 ± 11.55	948.90 ± 11.71
ECommerce	1505.86 ± 4.41	1513.19 ± 4.87	1598.01 ± 9.89	1598.00 ± 9.90	1484.18 ± 6.84	$\mathbf{1479.74 \pm 6.35}$
Traffic	5.389 ± 0.165	4.316 ± 0.071	$\mathbf{3.627 \pm 0.026}$	3.673 ± 0.018	4.117 ± 0.053	4.136 ± 0.058
Wiki2	3187.99 ± 43.73	2551.86 ± 8.65	2799.51 ± 6.55	2897.82 ± 16.20	$\mathbf{2465.16 \pm 2.57}$	2484.33 ± 7.77
Labour	22.73 ± 0.26	23.38 ± 0.13	$\mathbf{12.77 \pm 0.26}$	12.93 ± 0.38	15.03 ± 0.06	14.98 ± 0.06
M5	452.22 ± 2.62	442.05 ± 3.43	395.39 ± 8.18	$\mathbf{388.56 \pm 8.14}$	443.35 ± 2.45	442.25 ± 2.46
AEMO	730.73 ± 12.72	750.55 ± 10.24	$\mathbf{469.56 \pm 19.43}$	472.43 ± 17.76	635.85 ± 5.02	636.05 ± 5.22

Table S4: Continuous RMSE with PatchTST.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1147.15 ± 33.43	1378.98 ± 43.07	1187.81 ± 84.33	1189.75 ± 85.24	944.70 ± 34.42	$\mathbf{942.76 \pm 34.94}$
ECommerce	1621.20 ± 1.61	1613.68 ± 8.27	1695.95 ± 18.74	1695.95 ± 18.74	1580.75 ± 4.06	$\mathbf{1574.55 \pm 3.64}$
Traffic	5.398 ± 0.147	3.911 ± 0.017	3.819 ± 0.164	$\mathbf{3.733 \pm 0.116}$	3.774 ± 0.030	3.779 ± 0.030
Wiki2	3734.39 ± 56.08	3114.65 ± 18.45	3422.72 ± 38.18	3479.11 ± 42.90	$\mathbf{2938.79 \pm 9.03}$	2979.91 ± 24.03
Labour	24.55 ± 0.53	21.82 ± 0.84	15.56 ± 0.39	15.81 ± 0.44	$\mathbf{15.17 \pm 0.04}$	15.18 ± 0.04
M5	436.31 ± 2.45	443.77 ± 3.05	421.49 ± 27.55	$\mathbf{412.29 \pm 25.76}$	425.75 ± 0.96	423.21 ± 0.95
AEMO	649.84 ± 63.09	670.59 ± 62.21	$\mathbf{476.70 \pm 26.28}$	484.32 ± 26.61	582.29 ± 57.32	582.99 ± 57.93

Table S5: Continuous RMSE with DLinear.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1543.30 ± 34.15	1930.36 ± 17.26	1342.15 ± 122.72	1345.52 ± 123.83	$\mathbf{1111.47 \pm 19.09}$	1115.17 ± 22.23
ECommerce	1676.50 ± 5.92	1656.00 ± 6.57	1731.23 ± 46.59	1731.23 ± 46.59	1594.12 ± 2.48	$\mathbf{1586.34 \pm 2.54}$
Traffic	5.464 ± 0.078	4.195 ± 0.054	3.925 ± 0.155	$\mathbf{3.915 \pm 0.122}$	3.969 ± 0.042	3.974 ± 0.043
Wiki2	3988.23 ± 68.02	3296.17 ± 19.19	3572.01 ± 73.84	3647.37 ± 64.29	$\mathbf{3026.51 \pm 10.59}$	3077.44 ± 6.35
Labour	26.18 ± 0.13	24.31 ± 0.60	18.76 ± 0.54	18.54 ± 0.54	$\mathbf{16.90 \pm 0.20}$	$\mathbf{16.90 \pm 0.19}$
M5	474.85 ± 18.05	476.06 ± 11.93	433.42 ± 29.15	$\mathbf{425.61 \pm 26.70}$	451.99 ± 10.45	450.34 ± 9.78
AEMO	1194.62 ± 13.64	1205.63 ± 17.42	651.77 ± 19.69	$\mathbf{650.30 \pm 19.93}$	667.00 ± 5.75	666.93 ± 5.77

Table S6: Continuous RMSE with DUET.

Dataset	ProCAST	Ada-Y	PIR	Coherent PIR	Cov	ML
RH	1222.17 ± 13.95	1556.44 ± 35.57	992.27 ± 51.91	994.56 ± 53.74	911.36 ± 14.45	$\mathbf{910.77 \pm 14.93}$
ECommerce	1617.80 ± 2.40	1627.06 ± 3.36	1713.47 ± 11.59	1713.46 ± 11.60	1588.26 ± 5.99	$\mathbf{1582.03 \pm 4.18}$
Traffic	5.483 ± 0.159	4.379 ± 0.071	$\mathbf{3.695 \pm 0.029}$	3.740 ± 0.021	4.105 ± 0.046	4.121 ± 0.053
Wiki2	3759.16 ± 49.87	3122.88 ± 10.45	3458.26 ± 6.25	3509.17 ± 20.94	$\mathbf{2935.18 \pm 2.32}$	2968.11 ± 1.07
Labour	23.75 ± 0.22	23.86 ± 0.14	$\mathbf{13.37 \pm 0.23}$	13.55 ± 0.30	14.90 ± 0.06	14.81 ± 0.05
M5	458.64 ± 2.49	449.06 ± 3.53	400.87 ± 9.39	$\mathbf{393.70 \pm 9.29}$	442.69 ± 2.04	441.68 ± 2.03
AEMO	732.08 ± 13.04	751.69 ± 10.20	$\mathbf{473.61 \pm 21.97}$	476.27 ± 20.19	556.12 ± 8.32	555.46 ± 8.56

S2 Matched publication rules

PIR is the direct postprocessor output. Coherent PIR adds the shared nonnegative reconciliation; Protected PIR adds CPOR full projection. Each Coherent PIR target is constructed once per stream. Common, two-group shrinkage and Protected PIR use that same target, weights, reference, tolerance and feedback. The tables report mean RMSE and training-run SD, including cases where full projection is worse on ECommerce.

Table S7: Window publication of identical Coherent PIR candidates.

Dataset	Backbone	Coherent PIR	Common shrinkage	Two-group shrinkage	Protected PIR
RH	PatchTST	**1077.37 ± 55.71**	1152.43 ± 26.70	1149.13 ± 27.46	1116.47 ± 28.09
	DLinear	**1241.07 ± 107.86**	1459.72 ± 40.63	1453.81 ± 39.20	1421.91 ± 50.33
	DUET	**948.40 ± 38.12**	1152.64 ± 14.43	1126.11 ± 19.91	1089.05 ± 38.41
ECommerce	PatchTST	1582.66 ± 15.89	**1497.77 ± 6.36**	1497.85 ± 5.49	1498.49 ± 5.97
	DLinear	1617.31 ± 41.77	1541.91 ± 11.61	1541.55 ± 12.78	**1537.69 ± 17.46**
	DUET	1598.00 ± 9.90	1507.72 ± 4.57	**1507.56 ± 4.36**	1508.94 ± 4.00
Traffic	PatchTST	**3.660 ± 0.114**	3.807 ± 0.033	3.801 ± 0.033	3.749 ± 0.035
	DLinear	**3.855 ± 0.120**	4.086 ± 0.043	4.078 ± 0.043	3.988 ± 0.040
	DUET	**3.673 ± 0.018**	4.239 ± 0.065	4.229 ± 0.065	4.038 ± 0.049
Wiki2	PatchTST	2872.75 ± 37.17	2553.17 ± 8.27	2552.53 ± 8.16	**2512.58 ± 6.73**
	DLinear	3004.38 ± 51.77	2705.11 ± 15.14	2704.52 ± 15.18	**2643.44 ± 15.58**
	DUET	2897.82 ± 16.20	2558.81 ± 6.50	2558.23 ± 6.19	**2522.99 ± 1.21**
Labour	PatchTST	15.46 ± 0.42	15.70 ± 0.07	15.67 ± 0.09	**15.36 ± 0.11**
	DLinear	**18.37 ± 0.56**	19.22 ± 0.55	19.17 ± 0.54	18.75 ± 0.45
	DUET	**12.93 ± 0.38**	16.27 ± 0.08	16.18 ± 0.09	15.32 ± 0.07
M5	PatchTST	**408.00 ± 26.94**	426.31 ± 2.19	425.16 ± 2.78	417.60 ± 4.49
	DLinear	**419.79 ± 27.30**	453.94 ± 11.67	452.35 ± 11.21	440.85 ± 11.32
	DUET	**388.56 ± 8.14**	447.90 ± 3.30	446.18 ± 3.66	428.49 ± 5.83
AEMO	PatchTST	**481.64 ± 25.94**	693.34 ± 63.13	680.09 ± 62.58	654.12 ± 60.88
	DLinear	**645.09 ± 18.42**	995.27 ± 12.99	923.19 ± 9.39	866.66 ± 11.74
	DUET	**472.43 ± 17.76**	738.35 ± 3.89	708.51 ± 4.86	655.00 ± 4.94

S2.1 A same-state illustration

In the late-window Labour/PatchTST example with a reconciled PIR candidate, both rules released the reference at the first issue, so their allowances at the second issue are identical. One female part-time employment series in South Australia limits common shrinkage to 0.295% of the desired target move. Western Australia's regional-total correction is 4.552; common shrinkage retains 0.013 and full projection retains 0.073. Full projection therefore retains more of Western Australia's requested correction. Whether that correction reduces prediction error depends on the observation that follows.

Table S8: Continuous publication of identical Coherent PIR candidates.

Dataset	Backbone	Coherent PIR	Common shrinkage	Two-group shrinkage	Protected PIR
RH	PatchTST	1189.75 ± 85.24	1227.26 ± 36.86	1220.17 ± 37.00	$\mathbf{1187.20 \pm 39.78}$
	DLinear	$\mathbf{1345.52 \pm 123.83}$	1502.63 ± 73.13	1495.85 ± 70.83	1464.08 ± 83.25
	DUET	$\mathbf{994.56 \pm 53.74}$	1195.81 ± 18.22	1165.87 ± 26.64	1130.04 ± 49.52
ECommerce	PatchTST	1695.95 ± 18.74	$\mathbf{1610.14 \pm 6.70}$	1610.38 ± 6.85	1610.79 ± 7.87
	DLinear	1731.23 ± 46.59	1652.88 ± 17.42	1652.57 ± 16.66	$\mathbf{1649.37 \pm 22.22}$
	DUET	1713.46 ± 11.60	$\mathbf{1620.57 \pm 3.96}$	1621.28 ± 3.39	1623.08 ± 2.77
Traffic	PatchTST	$\mathbf{3.733 \pm 0.116}$	3.858 ± 0.033	3.848 ± 0.033	3.769 ± 0.038
	DLinear	$\mathbf{3.915 \pm 0.122}$	4.126 ± 0.049	4.111 ± 0.049	3.973 ± 0.051
	DUET	$\mathbf{3.740 \pm 0.021}$	4.257 ± 0.061	4.241 ± 0.058	3.974 ± 0.045
Wiki2	PatchTST	3479.11 ± 42.90	3115.94 ± 12.55	3115.17 ± 12.35	$\mathbf{3054.27 \pm 7.23}$
	DLinear	3647.37 ± 64.29	3293.52 ± 14.32	3292.59 ± 14.13	$\mathbf{3179.33 \pm 38.63}$
	DUET	3509.17 ± 20.94	3125.98 ± 7.32	3125.36 ± 6.78	$\mathbf{3067.23 \pm 7.33}$
Labour	PatchTST	$\mathbf{15.81 \pm 0.44}$	16.47 ± 0.08	16.43 ± 0.09	16.08 ± 0.15
	DLinear	$\mathbf{18.54 \pm 0.54}$	19.52 ± 0.49	19.44 ± 0.50	18.95 ± 0.38
	DUET	$\mathbf{13.55 \pm 0.30}$	16.57 ± 0.07	16.42 ± 0.10	15.42 ± 0.04
M5	PatchTST	$\mathbf{412.29 \pm 25.76}$	429.48 ± 4.81	427.71 ± 5.91	417.60 ± 7.98
	DLinear	$\mathbf{425.61 \pm 26.70}$	457.45 ± 14.18	454.96 ± 13.94	440.90 ± 13.39
	DUET	$\mathbf{393.70 \pm 9.29}$	446.78 ± 4.53	444.53 ± 4.73	417.00 ± 6.11
AEMO	PatchTST	$\mathbf{484.32 \pm 26.61}$	682.55 ± 64.78	661.38 ± 62.04	623.23 ± 58.26
	DLinear	$\mathbf{650.30 \pm 19.93}$	895.01 ± 7.28	834.79 ± 7.87	802.87 ± 13.55
	DUET	$\mathbf{476.27 \pm 20.19}$	700.52 ± 3.95	656.92 ± 2.62	591.43 ± 3.85

S3 Continuous allowances and levelwise effects

The control uses the same early calibration, variance weights, target selection and reference as the continuous rule. It resets only the remaining allowance before release 73. The continuous rule retains the allowance accumulated from the original start. Table S9 reports their 144-step RMSEs.

Table S9: Matched allowance reset and carry paths.

Dataset	Backbone	Cov reset	Cov carry	ML reset	ML carry
RH	PatchTST	1000.27 ± 31.30	944.70 ± 34.42	998.08 ± 31.81	942.76 ± 34.94
	DLinear	1234.00 ± 10.61	1111.47 ± 19.09	1236.71 ± 14.88	1115.17 ± 22.23
	DUET	990.12 ± 13.89	911.36 ± 14.45	987.21 ± 14.09	910.77 ± 14.93
ECommerce	PatchTST	1583.36 ± 4.46	1580.75 ± 4.06	1579.38 ± 4.09	1574.55 ± 3.64
	DLinear	1600.00 ± 2.82	1594.12 ± 2.48	1593.85 ± 3.83	1586.34 ± 2.54
	DUET	1591.55 ± 6.18	1588.26 ± 5.99	1587.39 ± 5.43	1582.03 ± 4.18
Traffic	PatchTST	3.802 ± 0.029	3.774 ± 0.030	3.807 ± 0.031	3.779 ± 0.030
	DLinear	4.035 ± 0.036	3.969 ± 0.042	4.039 ± 0.039	3.974 ± 0.043
	DUET	4.173 ± 0.053	4.105 ± 0.046	4.187 ± 0.059	4.121 ± 0.053
Wiki2	PatchTST	3010.17 ± 8.69	2938.79 ± 9.03	3032.58 ± 17.40	2979.91 ± 24.03
	DLinear	3119.61 ± 7.57	3026.51 ± 10.59	3143.58 ± 5.52	3077.44 ± 6.35
	DUET	3013.03 ± 3.17	2935.18 ± 2.32	3041.48 ± 1.54	2968.11 ± 1.07
Labour	PatchTST	15.67 ± 0.05	15.17 ± 0.04	15.68 ± 0.05	15.18 ± 0.04
	DLinear	18.19 ± 0.30	16.90 ± 0.20	18.18 ± 0.31	16.90 ± 0.19
	DUET	15.66 ± 0.06	14.90 ± 0.06	15.59 ± 0.06	14.81 ± 0.05
M5	PatchTST	427.44 ± 0.96	425.75 ± 0.96	425.72 ± 0.70	423.21 ± 0.95
	DLinear	454.85 ± 10.68	451.99 ± 10.45	453.70 ± 10.42	450.34 ± 9.78
	DUET	448.53 ± 2.47	442.69 ± 2.04	447.47 ± 2.40	441.68 ± 2.03
AEMO	PatchTST	631.80 ± 61.64	582.29 ± 57.32	632.46 ± 62.15	582.99 ± 57.93
	DLinear	811.19 ± 5.66	667.00 ± 5.75	811.13 ± 5.71	666.93 ± 5.77
	DUET	637.49 ± 5.29	556.12 ± 8.32	637.59 ± 5.38	555.46 ± 8.56

If a new zero-allowance contract is imposed at the midpoint, 3,304/4,806 Cov and 3,326/4,806

ML node sequences violate that additional limit at least once. These midpoint-start checks use a numerical ratio tolerance. The original-start envelope and cumulative checks use exact arithmetic and all pass. The two checks concern different starting times.

S3.1 Regional, bottom and intermediate levels

Tables S10 and S11 compare labelled regions and observed bottom levels with Protected PIR and NNLS. System totals are in the main paper. Other observed intermediate nodes contribute to the all-node metric.

Table S10: Window regional and bottom-level RMSE.

Level	Backbone	NNLS	Protected PIR	CPOR-Cov	CPOR-ML
Labour regions	PatchTST	23.08 ± 0.11	22.34 ± 0.12	$\mathbf{21.71 \pm 0.11}$	21.74 ± 0.12
	DLinear	30.62 ± 1.01	29.43 ± 0.81	$\mathbf{27.67 \pm 0.61}$	27.68 ± 0.61
	DUET	25.61 ± 0.19	23.31 ± 0.13	22.95 ± 0.12	$\mathbf{22.86 \pm 0.11}$
AEMO regions	PatchTST	454.23 ± 34.62	422.24 ± 32.64	399.79 ± 33.15	$\mathbf{399.78 \pm 33.61}$
	DLinear	746.41 ± 11.40	530.80 ± 7.17	492.53 ± 3.13	$\mathbf{492.51 \pm 3.16}$
	DUET	489.15 ± 3.98	410.18 ± 3.98	$\mathbf{396.37 \pm 3.51}$	396.41 ± 3.65
Traffic bottom	PatchTST	1.951 ± 0.003	1.891 ± 0.012	$\mathbf{1.862 \pm 0.003}$	1.865 ± 0.005
	DLinear	2.145 ± 0.007	2.040 ± 0.016	$\mathbf{2.004 \pm 0.007}$	2.007 ± 0.008
	DUET	2.019 ± 0.021	1.918 ± 0.005	$\mathbf{1.888 \pm 0.012}$	1.896 ± 0.012
Wiki2 bottom	PatchTST	1042.84 ± 3.41	1021.87 ± 3.35	$\mathbf{998.77 \pm 2.12}$	1003.51 ± 2.52
	DLinear	1074.76 ± 7.91	1049.54 ± 6.82	$\mathbf{1007.13 \pm 3.14}$	1009.64 ± 5.62
	DUET	1038.73 ± 2.72	1020.54 ± 1.87	$\mathbf{993.10 \pm 1.41}$	999.14 ± 3.49
Labour bottom	PatchTST	11.34 ± 0.02	11.00 ± 0.11	10.45 ± 0.01	$\mathbf{10.43 \pm 0.01}$
	DLinear	12.60 ± 0.28	12.09 ± 0.20	$\mathbf{11.45 \pm 0.18}$	$\mathbf{11.45 \pm 0.19}$
	DUET	10.79 ± 0.03	10.18 ± 0.06	9.89 ± 0.02	$\mathbf{9.88 \pm 0.02}$
M5 bottom	PatchTST	192.69 ± 0.26	$\mathbf{187.14 \pm 1.58}$	189.37 ± 0.21	188.70 ± 0.17
	DLinear	203.64 ± 3.69	$\mathbf{195.68 \pm 3.61}$	198.82 ± 3.63	198.19 ± 3.49
	DUET	200.30 ± 1.05	$\mathbf{188.94 \pm 2.05}$	195.01 ± 0.82	194.38 ± 0.80

Three-run mean ± sample SD. Boldface marks the lowest displayed mean within each row. Protected PIR and Cov/ML use the same protection rule with different targets and their own allowances.

Table S11: Continuous regional and bottom-level RMSE.

Level	Backbone	NNLS	Protected PIR	CPOR-Cov	CPOR-ML
Labour regions	PatchTST	24.99 ± 0.08	23.86 ± 0.12	$\mathbf{22.90 \pm 0.08}$	22.99 ± 0.07
	DLinear	31.56 ± 0.91	29.99 ± 0.67	$\mathbf{26.57 \pm 0.26}$	26.60 ± 0.24
	DUET	27.43 ± 0.17	23.76 ± 0.07	23.05 ± 0.09	$\mathbf{22.91 \pm 0.09}$
AEMO regions	PatchTST	455.73 ± 34.41	405.13 ± 30.77	371.26 ± 31.04	$\mathbf{370.88 \pm 31.73}$
	DLinear	761.38 ± 11.62	492.72 ± 8.20	407.65 ± 3.07	$\mathbf{407.61 \pm 3.08}$
	DUET	491.12 ± 3.99	375.53 ± 3.56	349.46 ± 5.26	$\mathbf{348.89 \pm 5.52}$
Traffic bottom	PatchTST	2.022 ± 0.002	1.935 ± 0.018	$\mathbf{1.897 \pm 0.004}$	1.900 ± 0.004
	DLinear	2.208 ± 0.007	2.055 ± 0.028	$\mathbf{1.988 \pm 0.008}$	1.991 ± 0.008
	DUET	2.083 ± 0.020	1.939 ± 0.003	$\mathbf{1.895 \pm 0.004}$	1.902 ± 0.006
Wiki2 bottom	PatchTST	1324.59 ± 5.26	1292.20 ± 4.73	$\mathbf{1238.92 \pm 3.88}$	1250.42 ± 8.28
	DLinear	1370.25 ± 8.54	1324.92 ± 13.71	$\mathbf{1256.14 \pm 4.57}$	1278.26 ± 2.98
	DUET	1320.90 ± 2.89	1292.04 ± 0.92	$\mathbf{1234.09 \pm 0.75}$	1244.78 ± 0.86
Labour bottom	PatchTST	11.66 ± 0.01	11.20 ± 0.15	10.25 ± 0.01	$\mathbf{10.21 \pm 0.01}$
	DLinear	12.80 ± 0.28	12.08 ± 0.19	10.81 ± 0.13	$\mathbf{10.80 \pm 0.13}$
	DUET	11.03 ± 0.03	10.08 ± 0.08	9.62 ± 0.03	$\mathbf{9.58 \pm 0.03}$
M5 bottom	PatchTST	194.79 ± 0.28	$\mathbf{186.29 \pm 2.91}$	190.65 ± 0.25	189.06 ± 0.19
	DLinear	206.01 ± 3.58	$\mathbf{194.23 \pm 4.51}$	199.48 ± 3.52	198.29 ± 3.12
	DUET	202.17 ± 1.08	$\mathbf{184.45 \pm 2.12}$	194.77 ± 0.71	193.71 ± 0.70

Three-run mean ± sample SD. Boldface marks the lowest displayed mean within each row. Protected PIR and Cov/ML use the same protection rule with different targets and their own allowances.

S4 Dataset-cluster uncertainty

For the window protocol, we first average the two window RMSEs within each backbone and run, then compute the paired percentage reduction. For the continuous protocol, we use RMSE computed over all 144 observations. For either protocol, averaging paired reductions over the three backbones and three runs gives one effect per dataset. We enumerate all $7^7 = 823{,}543$ bootstrap resamples and take linear-interpolated 2.5th and 97.5th percentiles. These intervals resample the selected datasets, treating them as exchangeable. They describe uncertainty in this retrospective comparison. A two-sided sign-flip test enumerates all $2^7 = 128$ signs and assumes independent, symmetric dataset effects under its null. Training repetitions are not separate dataset clusters.

Holm adjustment is applied to six complete-procedure contrasts per protocol (Cov/ML against NNLS, ProCAST and PIR). The two matched-PIR contrasts form a separate family within each protocol.

Table S12: Complete-procedure paired effects.

Protocol	Method	Comparator	Gain (%)	Empirical 95% interval	p	Holm p
Window	Cov	NNLS	9.32	[3.57, 15.82]	0.0156	0.0938
Window	Cov	ProCAST	17.13	[9.26, 25.01]	0.0156	0.0938
Window	Cov	PIR	-3.21	[-13.73, 5.20]	0.7031	1.0000
Window	ML	NNLS	9.33	[3.66, 15.86]	0.0156	0.0938
Window	ML	ProCAST	17.16	[9.40, 25.02]	0.0156	0.0938
Window	ML	PIR	-3.19	[-13.82, 5.24]	0.6875	1.0000
Continuous	Cov	NNLS	13.20	[5.12, 22.38]	0.0156	0.0938
Continuous	Cov	ProCAST	20.49	[11.76, 28.74]	0.0156	0.0938
Continuous	Cov	PIR	2.04	[-5.39, 9.56]	0.6406	1.0000
Continuous	ML	NNLS	13.11	[5.16, 22.24]	0.0156	0.0938
Continuous	ML	ProCAST	20.44	[11.87, 28.58]	0.0156	0.0938
Continuous	ML	PIR	1.98	[-5.32, 9.19]	0.6406	1.0000

Table S13: Matched PIR release effects.

Protocol	Method	Comparator	Gain (%)	Empirical 95% interval	p	Holm p
Window	Protected PIR	Common shrinkage	3.57	[1.77, 5.96]	0.0156	0.0312
Window	Protected PIR	Two-group shrinkage	2.72	[1.56, 3.98]	0.0156	0.0312
Continuous	Protected PIR	Common shrinkage	4.34	[2.23, 7.07]	0.0156	0.0312
Continuous	Protected PIR	Two-group shrinkage	3.28	[1.94, 4.62]	0.0156	0.0312

Even if all seven dataset effects have the same sign, the smallest two-sided sign-flip p-value is $2/128 = 0.015625$. This exceeds the first Holm threshold, $0.05/6$, for the six complete-procedure comparisons. The matched-PIR family compares Protected PIR with common and two-group shrinkage of the same Coherent PIR candidate. This family contains only these two release-rule comparisons. Cov/ML versus Protected PIR is a different comparison. The tests use the selected datasets and dates; performance on new datasets or periods requires further observations. The matched-PIR mean reduction remains positive after leaving out any one dataset.

S5 AEMO whole-month extension

The monthly comparison uses all 1,488 October half-hours of estimated underlying demand at five regions and their system total. All methods use the same finalized observations, three backbones and three training runs. Each protected path starts with zero allowance and retains it throughout the month. The locally recorded outcome-specific protocol preceded data acquisition; later backbone and short-window extensions are retrospective. Feedback is assumed to arrive before the next release, so this evaluation does not reconstruct historical reporting delays or revisions.

Table S14: Complete AEMO monthly comparison.

Method	PatchTST	DLinear	DUET
Ada-Y	672.74 ± 34.59	1198.84 ± 14.54	743.40 ± 18.94
CPOR-Cov	291.10 ± 13.19	288.59 ± 2.53	252.37 ± 2.33
CPOR-ML	286.47 ± 13.27	287.82 ± 2.23	248.10 ± 2.04
Fixed expert	412.91 ± 20.01	197.22 ± 0.00	331.62 ± 116.39
Cov common shrinkage	318.09 ± 14.53	319.11 ± 6.15	276.13 ± 1.75
Unprotected Cov	190.23 ± 0.48	198.40 ± 0.47	200.82 ± 0.58
ML common shrinkage	313.24 ± 14.49	318.61 ± 6.15	271.22 ± 1.60
Unprotected ML	183.22 ± 0.63	197.22 ± 0.00	195.53 ± 0.93
NNLS reference	667.19 ± 29.72	1234.14 ± 17.39	757.24 ± 16.46
PIR	609.65 ± 26.68	709.37 ± 4.78	480.91 ± 4.97
Coherent PIR	601.08 ± 27.74	707.74 ± 4.44	469.39 ± 5.45
Common shrinkage (PIR)	604.94 ± 31.89	728.34 ± 4.14	489.66 ± 3.27
Protected PIR	569.58 ± 30.11	719.98 ± 4.08	477.74 ± 6.06
Two-group shrinkage (PIR)	585.69 ± 29.25	725.22 ± 4.29	482.16 ± 4.93
ProCAST	614.34 ± 29.16	1174.72 ± 14.22	698.82 ± 12.89
Protected Ada-Y	661.11 ± 31.69	1198.15 ± 15.08	739.26 ± 19.99
Protected fixed expert	425.74 ± 21.32	287.82 ± 2.23	352.84 ± 91.79
Protected MinT (Cov)	293.54 ± 12.11	288.62 ± 2.54	252.15 ± 2.33
Protected MinT (raw)	658.38 ± 27.45	1234.13 ± 17.39	752.16 ± 16.53
Protected ProCAST	614.63 ± 29.16	1172.26 ± 13.70	698.89 ± 12.88
Protected rolling MinT	648.19 ± 30.74	1234.13 ± 17.39	726.51 ± 21.99
Protected WLS (raw)	663.58 ± 29.92	1234.13 ± 17.39	751.07 ± 17.92
Base forecast	669.44 ± 30.40	1234.14 ± 17.39	755.22 ± 18.11

Table S15: AEMO system and regional RMSE.

Node	NNLS	Protected PIR	Cov common	CPOR-Cov	ML common	CPOR-ML
PatchTST						
System	1704.85 ± 104.71	1343.59 ± 114.25	807.61 ± 50.60	733.06 ± 47.34	797.33 ± 50.40	$\mathbf{723.47 \pm 47.59}$
NSW	588.06 ± 60.52	558.95 ± 54.47	312.06 ± 23.54	289.20 ± 16.33	304.51 ± 23.72	$\mathbf{282.96 \pm 15.83}$
QLD	329.96 ± 7.35	327.81 ± 7.53	202.52 ± 2.44	210.15 ± 13.17	$\mathbf{196.34 \pm 3.44}$	203.23 ± 14.19
SA	327.98 ± 3.97	282.03 ± 6.45	132.39 ± 3.67	81.15 ± 1.98	130.12 ± 3.09	$\mathbf{79.94 \pm 2.00}$
TAS	95.72 ± 2.55	85.21 ± 0.52	40.81 ± 1.92	33.71 ± 1.42	39.69 ± 1.85	$\mathbf{32.81 \pm 1.35}$
VIC	956.57 ± 25.33	819.91 ± 24.69	413.16 ± 13.28	399.31 ± 10.52	411.45 ± 13.09	$\mathbf{396.44 \pm 10.70}$
DLinear						
System	3636.33 ± 49.70	2078.95 ± 15.44	927.54 ± 18.89	832.63 ± 7.65	926.35 ± 19.03	$\mathbf{830.59 \pm 6.86}$
NSW	1336.18 ± 18.24	790.96 ± 3.98	354.41 ± 7.08	313.79 ± 2.63	354.11 ± 7.15	$\mathbf{313.11 \pm 2.28}$
QLD	1136.70 ± 17.22	668.26 ± 4.06	285.16 ± 3.83	258.69 ± 1.09	284.80 ± 3.72	$\mathbf{258.22 \pm 1.45}$
SA	304.07 ± 4.84	199.86 ± 0.60	77.50 ± 0.70	73.78 ± 0.50	77.29 ± 0.84	$\mathbf{73.55 \pm 0.59}$
TAS	80.44 ± 1.25	55.46 ± 0.69	27.80 ± 0.22	27.15 ± 0.21	27.24 ± 0.22	$\mathbf{26.58 \pm 0.21}$
VIC	911.11 ± 15.73	526.41 ± 7.70	242.26 ± 6.22	225.51 ± 4.51	241.83 ± 6.07	$\mathbf{224.84 \pm 4.17}$
DUET						
System	2122.64 ± 55.12	1205.00 ± 25.66	763.57 ± 6.90	688.80 ± 6.54	751.15 ± 7.11	$\mathbf{678.44 \pm 6.25}$
NSW	777.71 ± 27.81	458.34 ± 17.55	305.33 ± 6.46	278.44 ± 6.25	297.48 ± 6.93	$\mathbf{271.79 \pm 6.42}$
QLD	488.85 ± 27.30	348.20 ± 2.63	220.69 ± 3.04	212.67 ± 0.64	214.88 ± 2.79	$\mathbf{207.67 \pm 0.66}$
SA	273.47 ± 4.60	198.58 ± 5.48	91.69 ± 2.61	72.42 ± 0.70	90.42 ± 2.56	$\mathbf{71.88 \pm 0.66}$
TAS	80.58 ± 2.70	52.37 ± 0.65	32.10 ± 0.62	29.51 ± 0.58	31.22 ± 0.58	$\mathbf{28.61 \pm 0.55}$
VIC	800.21 ± 9.17	603.93 ± 19.90	243.41 ± 11.74	232.41 ± 9.69	242.21 ± 12.32	$\mathbf{230.20 \pm 9.73}$

Whole-month RMSE in MW, three-run mean ± SD. All listed procedures satisfy the same protection contract; boldface marks the lowest displayed mean per node and backbone.

Under DLinear, the protected fixed expert matches ML at the displayed precision, so this comparison does not show an additional benefit from online combination there. Nodewise results retain the PatchTST reversals in Queensland: both full projections have higher RMSE than their corresponding common-shrinkage rules despite lower overall error.

S6 Additional protection comparisons

These analyses use their original prediction sequences to examine the outcome domain and numerical procedure. Their evaluation grids differ from the seven-dataset, three-backbone main comparison.

S6.1 Nonnegative versus all-real envelopes

This comparison covers six benchmarks, two backbones, two windows and three runs. The rules use the same targets, reference, variance weights and feedback. Only the allowed observation domain changes; each rule updates its own allowance from the forecast it releases.

Table S16: Outcome-domain comparison across three training runs.

Dataset	Backbone	Cov	Cov all-real	ML	ML all-real
RH	PatchTST	948.663 ± 23.189	987.649 ± 24.719	947.268 ± 23.815	987.162 ± 25.333
	DLinear	1192.969 ± 5.593	1355.521 ± 7.602	1193.892 ± 8.111	1355.632 ± 7.944
ECommerce	PatchTST	1476.194 ± 4.579	1480.250 ± 4.981	1470.642 ± 3.516	1477.222 ± 4.626
	DLinear	1494.238 ± 3.465	1510.508 ± 3.592	1484.681 ± 2.453	1507.059 ± 3.104
Traffic	PatchTST	3.739 ± 0.030	3.764 ± 0.029	3.743 ± 0.032	3.765 ± 0.029
	DLinear	3.977 ± 0.037	3.995 ± 0.037	3.981 ± 0.040	4.001 ± 0.041
Wiki2	PatchTST	2465.831 ± 5.546	2506.700 ± 8.077	2483.485 ± 8.019	2509.814 ± 8.806
	DLinear	2548.730 ± 4.698	2625.325 ± 11.335	2557.009 ± 9.661	2629.790 ± 11.225
Labour	PatchTST	14.808 ± 0.056	14.813 ± 0.058	14.812 ± 0.059	14.819 ± 0.059
	DLinear	17.678 ± 0.356	17.676 ± 0.357	17.685 ± 0.359	17.684 ± 0.360
M5	PatchTST	421.624 ± 0.571	422.660 ± 0.539	420.117 ± 0.463	421.200 ± 0.486
	DLinear	447.254 ± 10.573	449.185 ± 10.833	445.953 ± 10.235	447.784 ± 10.481

S6.2 Direct and two-stage protection

Let a_t be the raw combined target and z_t its variance-weighted nonnegative reconciliation. We compare protecting a_t directly with protecting z_t, using the same reference, variance weights, 0.5% tolerance and implementation-specific numerical routine. Each rule starts with zero allowance and updates it using its own released errors. Eq. (24) in the main paper projects the raw combined target directly. The two-stage rule first reconciles that target, then projects the reconciled forecast. Both use the numerical release checks. The two orders can give different forecasts because they minimize distances to different targets.

This historical comparison uses the original six datasets and two backbones with three training runs, including only Labour's late window. Both projection orders satisfy the nodewise cumulative limits in all tested cases.

On the evaluated streams, direct projection gives mean gains of 0.0116% and 0.0277% on the four observed bottom levels, but mean system-total changes of -0.0136% and -0.0439%. The strongest overall effects favor direct projection on Wiki2 and two-stage projection on RH.

Table S17: Direct versus two-stage protection: paired RMSE reductions (%).

Dataset / level	Cov	ML
ECommerce	0.000000	0.000000
Labour	0.000000	0.000000
M5	0.000000	0.000000
RH	-0.035494	-0.017580
Traffic	-0.000336	0.000000
Wiki2	+0.052765	+0.083512
Equal-weight mean	+0.002822	+0.010989
Labour System	0.000000	0.000000
M5 System	0.000000	0.000000
RH System	-0.008824	-0.029121
Traffic System	-0.000014	0.000000
Wiki2 System	-0.059153	-0.190450
Labour Bottom	0.000000	0.000000
M5 Bottom	0.000000	0.000000
Traffic Bottom	-0.000655	0.000000
Wiki2 Bottom	+0.046886	+0.110741
Labour Region	0.000000	0.000000
M5 State	0.000000	0.000000

Positive values favor direct protection. Windows are averaged within each dataset, backbone and run before calculating paired changes. Values below 0.0000005% in magnitude are displayed as zero.

S7 Computational cost and numerical events

The numerical checks and timing study use the original six datasets and two backbones, with only Labour's late window. They measure these procedures separately from the expanded-grid runs.

Table S18: Observed publication events and coherence residuals.

			Releases with an event			Coherence residual	
Dataset	Impl.	Releases	Retry	Clip	Guard	P95	Maximum
RH	Cov	864	0	156	0	2.91e-11	1.46e-08
RH	ML	864	0	160	0	3.27e-11	1.33e-07
ECommerce	Cov	864	0	40	0	2.27e-13	6.32e-11
ECommerce	ML	864	0	28	0	0.00e+00	1.32e-11
Traffic	Cov	864	0	533	0	3.33e-11	2.17e-08
Traffic	ML	864	0	540	0	2.97e-11	7.85e-09
Wiki2	Cov	864	136	606	0	7.69e-08	1.51e-06
Wiki2	ML	864	141	641	0	5.76e-08	2.05e-06
Labour	Cov	432	0	139	0	3.21e-08	2.89e-07
Labour	ML	432	0	151	0	3.96e-08	1.62e-07
M5	Cov	864	0	0	106	0.00e+00	2.91e-11
M5	ML	864	0	79	0	1.46e-11	2.60e-07

Retry: more than one projection attempt; Clip: a coordinate repair; Guard: a contraction toward the reference. Counts refer to releases and can overlap.

ML is faster in all twelve measured dataset–backbone groups. Cov's error-moment estimation and nodewise quadratic optimization account for most of the difference. Both implementations enforce the same cumulative limits. These measurements do not establish a DUET runtime ranking.

Table S19: Postprocessing time on the original benchmark windows.

			Target generation		Protected publication		Total	
Dataset	Backbone	CPOR	Mean	P95	Mean	P95	Mean	P95
RH	PatchTST	Cov	17.08	19.63	2.79	3.11	19.87	22.74
RH	PatchTST	ML	**3.09**	**3.75**	**1.49**	**1.59**	**4.58**	**5.25**
RH	DLinear	Cov	20.38	24.46	3.22	4.19	23.60	28.82
RH	DLinear	ML	**3.81**	**4.63**	**1.78**	**2.27**	**5.59**	**6.57**
ECommerce	PatchTST	Cov	23.97	25.66	3.54	3.68	27.51	29.22
ECommerce	PatchTST	ML	**5.93**	**6.56**	**2.04**	**2.22**	**7.97**	**8.77**
ECommerce	DLinear	Cov	25.17	29.87	3.72	5.17	28.89	34.54
ECommerce	DLinear	ML	**6.13**	**7.92**	**2.14**	**2.68**	**8.27**	**10.64**
Traffic	PatchTST	Cov	197.64	240.50	15.94	19.61	213.58	256.96
Traffic	PatchTST	ML	**99.66**	**130.44**	**12.10**	**17.04**	**111.76**	**144.92**
Traffic	DLinear	Cov	221.20	260.64	17.31	22.67	238.51	277.94
Traffic	DLinear	ML	**110.29**	**127.06**	**12.71**	**16.51**	**123.00**	**139.69**
Wiki2	PatchTST	Cov	214.28	242.06	22.80	56.55	237.08	273.89
Wiki2	PatchTST	ML	**104.58**	**121.12**	**19.24**	**51.76**	**123.82**	**156.21**
Wiki2	DLinear	Cov	213.26	238.86	19.45	**42.99**	232.71	264.37
Wiki2	DLinear	ML	**106.54**	**129.01**	**15.63**	46.53	**122.18**	**155.42**
Labour	PatchTST	Cov	65.36	77.12	5.78	7.53	71.14	83.81
Labour	PatchTST	ML	**28.88**	**39.18**	**4.29**	**5.36**	**33.17**	**42.96**
Labour	DLinear	Cov	69.91	82.62	5.83	7.66	75.73	88.63
Labour	DLinear	ML	**30.59**	**35.38**	**4.08**	**5.24**	**34.67**	**39.54**
M5	PatchTST	Cov	55.99	64.90	3.42	4.61	59.42	68.25
M5	PatchTST	ML	**21.72**	**26.73**	**3.31**	**4.57**	**25.03**	**30.53**
M5	DLinear	Cov	54.89	63.32	3.49	4.25	58.37	67.07
M5	DLinear	ML	**21.53**	**23.64**	**3.19**	**3.58**	**24.72**	**27.71**

Time in milliseconds, pooled across windows and three runs. P95 denotes the 95th percentile. Boldface marks lower costs within each dataset–backbone pair.

S8 Additional recent comparisons

The following comparisons use their original datasets, windows and training runs. They are summarized separately from the current 21-group main comparison.

S8.1 Fixed-stream adapters and robust reconciliation

This historical comparison uses six datasets, two backbones and the original 22 configurations, including only Labour late. Table S20 uses one initialization throughout. Percentages in the main text are computed within each dataset–backbone pair from unrounded RMSE values, then averaged with equal group weights.

Table S20: Mean-node RMSE on the first training run. P denotes the CPOR release rule.

Dataset	Backbone	COSA	COSA+P	Robust	Robust+P	Cov	ML
ECommerce	DLinear	1,547.28	1,546.72	7,805.10	1,541.78	1,496.66	1,486.73
ECommerce	PatchTST	1,505.86	1,503.09	6,408.82	1,505.74	1,479.70	1,473.83
Labour	DLinear	23.15	22.06	1,033.60	22.25	21.07	21.08
Labour	PatchTST	20.91	20.08	230.08	19.92	18.86	18.89
M5	DLinear	458.03	457.00	4,981.35	454.54	443.00	441.99
M5	PatchTST	431.80	430.72	10,821.92	422.75	422.15	420.64
RH	DLinear	1,630.78	1,620.79	4,260.55	1,622.45	1,194.92	1,200.63
RH	PatchTST	1,128.94	1,127.73	1,670.92	1,084.53	925.54	923.33
Traffic	DLinear	4.11	4.11	21.09	4.10	3.98	3.99
Traffic	PatchTST	3.85	3.86	39.41	3.84	3.77	3.77
Wiki2	DLinear	2,696.01	2,697.48	34,258.51	2,662.73	2,551.53	2,556.24
Wiki2	PatchTST	2,535.14	2,537.27	28,929.90	2,539.14	2,463.29	2,478.90

COSA. We adapt the authors' output-space correction to the same fixed forecasts and sequential feedback used by CPOR. Updates use only observations already returned. On the original ETTh1/DLinear task at horizon 96, the authors' driver with a newly trained backbone gave MSE 0.4548, compared with the published COSA-P value of 0.4482.

Robust reconciliation. Let X and Y contain calibration forecasts and observations as rows. Each day's observation vector lies in the column space of S, up to numerical precision. For a linear map B acting on forecast rows, define $B_0 = X^\dagger Y$, where $\dagger$ denotes the Moore–Penrose inverse. The normal equations $X^\top(Y - XB_0) = 0$ give

$$(Y - XB)^\top(Y - XB) = (Y - XB_0)^\top(Y - XB_0) + (B - B_0)^\top X^\top X(B - B_0). \tag{S1}$$

The added term is positive semidefinite. Hence B_0 minimizes empirical quadratic loss for every positive semidefinite precision matrix, and also its maximum over a nonempty set of such matrices. We use the full bootstrap uncertainty set, whose nonemptiness is checked. Its fitted forecasts are linearly coherent; CPOR additionally enforces nonnegativity and the cumulative error limits.

The minimum-norm map sends directions absent from calibration to zero. We also evaluate

$$B_1 = B_0 + (I - X^\dagger X)SS^\dagger, \tag{S2}$$

which preserves the same calibration fit while passing the coherent projection of those directions through unchanged. Cov and ML reduce RMSE relative to this protected alternative by 5.97% and 5.99% on average. Thus the poor direct results of the minimum-norm map depend on how it extrapolates beyond the calibration data. These maps use 36 calibration dates. On the authors' Australian births task with Prophet, our total and bottom-level relative RMSEs were 0.970 and 0.976, versus 0.968 and 0.973 in the original study. A separate semidefinite-program check gave the same objective with unrestricted fitted coordinates and with the least-squares solution fixed.

Cost and scope. We compared releasing each external target directly with projecting that same target under the aggregation, nonnegativity and cumulative error constraints. Relative to direct release, the joint projection reduced RMSE by a mean of 0.80% for COSA (median 0.15%) and 82.01% for the minimum-norm map (median 90.57%). The large change for the linear map reflects its poor direct-release accuracy on these streams. Both protected targets satisfy the cumulative limits; coherence is checked within numerical tolerance. Runtime records separate target computation from protected publication. Across the 1,584 scored issues, the measured COSA target and protection stages averaged 47.05 and 5.89 ms per issue. The minimum-norm algebraic fit and application together took 0.17 s over the 22 configurations; subsequent protection averaged 7.80 ms per issue. These are local elapsed times for the adapted implementations.

S8.2 Two-stage residual correction

Table S21 uses the same 22 configurations and first backbone training run as Section S8.1. Base denotes the supplied backbone forecasts. Direct releases the residual-corrected forecast. Coherent reconciles that same forecast, and Protected also applies the CPOR protection rule. Cov and ML use their own targets. Each group receives equal weight in paired percentage changes. The corrected target alone had lower RMSE than the raw forecasts in eight groups. Its largest deterioration occurred on RH/PatchTST; imposing the cumulative limit there brought its RMSE just below that of the raw forecasts.

Table S21: Mean-node RMSE by dataset and backbone for the residual corrector.

Dataset	Backbone	Base	Direct	Coherent	Protected	Cov	ML
ECommerce	DLinear	1,547.704	1,541.747	1,541.747	1,541.914	1,496.662	**1,486.729**
ECommerce	PatchTST	1,501.689	1,497.877	1,497.877	1,498.007	1,479.697	**1,473.829**
Labour	DLinear	22.354	22.215	22.202	22.224	**21.075**	21.084
Labour	PatchTST	20.174	20.209	20.264	20.111	**18.859**	18.888
M5	DLinear	456.043	456.355	456.384	455.989	443.003	**441.994**
M5	PatchTST	430.087	429.252	429.084	428.946	422.148	**420.644**
RH	DLinear	1,619.559	1,607.244	1,610.874	1,605.058	**1,194.925**	1,200.632
RH	PatchTST	1,128.575	1,140.044	1,140.852	1,127.655	925.544	**923.329**
Traffic	DLinear	4.121	4.101	4.095	4.098	**3.983**	3.987
Traffic	PatchTST	3.851	3.852	3.838	3.859	**3.767**	3.773
Wiki2	DLinear	2,700.155	2,696.618	2,696.462	2,696.447	**2,551.527**	2,556.235
Wiki2	PatchTST	2,534.776	2,531.499	2,561.951	2,534.091	**2,463.292**	2,478.903

The residual corrector is trained on pre-scoring forecast errors using the authors' Huber objective and selected on earlier validation data. Within each forecast block its target does not use unreturned observations. The protected arm uses that same target under CPOR's sequential rule.

Checks and disaggregation. On ETTh1 with an iTransformer and a 96-step input and horizon, the original publication reports corrected MSE/MAE of 0.367/0.311. Under the public code's residual-series time alignment and pooled normalization, we obtained 0.370/0.313. The same base predictions on those dates gave 0.369/0.312. Retaining common absolute split boundaries also produced a slight deterioration after correction. Correction did not improve accuracy in these reproduction checks. Our main comparison adapts the method to shorter horizons using fixed PatchTST and DLinear predictions.

The protected residual target satisfies the cumulative limits at every evaluated prefix. Relative to the protected residual target, Cov and ML had lower system-total RMSE in nine of the ten applicable groups. Traffic/DLinear was the exception: the protected residual target achieved 135.89 versus 136.61 and 136.73 for Cov and ML. On Labour, region-averaged RMSE decreased in both backbone groups, by means of 5.12% and 4.97%, respectively.

S8.3 MCGrad regression extension

We applied the authors' continuous-outcome extension of MCGrad to the first PatchTST and DLinear runs on the original 22 configurations. The published study evaluates predicted probabilities. The regression extension is fitted on earlier observations and selected using pre-scoring validation data. The fixed fit selected zero correction rounds in 16 of 22 configurations. A sensitivity run refitted the model before each subsequent forecast block, using only observations already returned.

Direct fixed and refreshed targets increased RMSE by 0.71% and 0.84%, respectively, relative to the raw forecasts. For each target, common shrinkage first enforces nonnegative coherence,

Table S22: Release comparisons for the MCGrad regression target.

	Fixed fit	Block refit
Direct target: sequences ever over limit (of 1,998)	1,385	1,542
Common shrinkage: sequences ever over limit	0	0
Full-set release: sequences ever over limit	0	0
Full-set RMSE reduction vs. shrinkage (%)	0.71	0.59
Groups improving (of 12)	9	11

while full-set release directly applies the joint constraints. The two rules use the same external target and cumulative limits, but apply coherence and protection in different orders. In the separate complete-procedure comparison against the protected fixed-fit target, Cov and ML reduced RMSE by 6.03% and 6.04%; those differences include target generation. Both fits use the same datasets and backbone initialization.

S8.4 Li et al. same-task preprint

We used Li et al.'s Algorithm 1 (arXiv:2608.13886v1), with separate bottom-series weights and ridge shrinkage toward equal weights (Separate + eRidge). Diag shrinks error covariance toward its diagonal; Factor uses the cross-series mean error as a one-factor target and is among the authors' strongest configurations. Both use the same candidates, including cross-branch forecasts. Ridge penalties are selected on earlier validation data and weights are refitted before each release. Each window has 72 scored forecasts from one backbone initialization. Factor selected the largest ridge penalty (10^5) in 173 of 1,152 releases: 109 on M5 and 64 on Wiki2.

Table S23: Mean-node RMSE on the common forecast streams.

Dataset	Backbone	Li (Diag)	Li (Factor)	CPOR-Cov	CPOR-ML
RH	PatchTST	—	—	925.544	923.329
	DLinear	—	—	1194.925	1200.632
ECommerce	PatchTST	—	—	1479.697	1473.829
	DLinear	—	—	1496.662	1486.729
Traffic	PatchTST	3.914	3.901	**3.767**	3.773
	DLinear	4.111	4.120	**3.983**	3.987
Wiki2	PatchTST	2543.930	2551.921	**2463.292**	2478.903
	DLinear	2700.139	2700.164	**2551.527**	2556.235
Labour	PatchTST	16.097	16.473	**14.865**	14.871
	DLinear	18.951	18.960	**17.295**	17.299
M5	PatchTST	429.861	433.889	422.148	**420.644**
	DLinear	456.024	456.072	443.003	**441.994**

Table S23 averages RMSE over windows for each backbone. RH and ECommerce lack the bottom-level inputs required by Li et al. For percentage comparisons we use the unrounded group means in the numerator and denominator, then average the eight group percentages with equal weight. Factor reduced mean-node RMSE relative to Diag in 2 of 16 windows and one of eight groups (Traffic/PatchTST); its group-averaged RMSE change was a 0.44% increase.

Relative to Factor, Cov and ML reduced system RMSE by 4.77% and 4.58%, intermediate-level RMSE by 4.38% and 4.32%, and bottom-level RMSE by 5.52% and 5.51%. The corresponding reductions relative to Diag were 4.86% and 4.67%, 3.95% and 3.88%, and 5.02% and 5.01%. Intermediate levels include all nonsystem, nonbottom nodes: in Labour, eight regions and sixteen sex-by-region series. For Traffic/DLinear's early window, Factor had lower system and intermediate-level RMSE than either CPOR implementation. In Wiki2/PatchTST's late window, ML increased bottom-level RMSE by 0.016% relative to Diag.